MarketMind: An AI-Powered Trading Assistant (2026)

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ABSTRACT

MarketMind is an AI-powered trading system designed to predict short-term stock direction using a combination of technical, fundamental, and macroeconomic indicators. The model uses a neural network with fuzzy activation functions to better capture uncertainty and nonlinear market behavior. Using a 20-day lookback window, the system predicts stock movement 3 days into the future. The fuzzy activation model achieved **60.9%** directional accuracy, outperforming a baseline ReLU network (53.9%) and random chance (50%). The system integrates predictions into a user-friendly interface to support clearer and more confident decision-making.

Data-Driven · AI-Powered · Better Decisions · Uncertainty Aware

INTRODUCTION / PROBLEM

- Stock prices are volatile and difficult to predict. This uncertainty deters many people from investing.
- Investors need a system that integrates technical, fundamental, and macroeconomic indicators for better buy/sell decisions in a simple to use interface.
- Traditional models often fail to capture nonlinear, uncertain, and time-dependent market behavior.

BACKGROUND

- **Traditional Models (ARIMA, GARCH)** Work with markets that were simpler and slower.
- **Machine Learning (RF, SVM)** Improved performance but struggle with nonlinear, time-based patterns.
- **Deep Learning (ANN, LSTM)** Captures sequences and complex trends in stock data.
- **Fuzzy Neural Networks** Outperform standard activations by ≈ 3–5% in MSE across most stocks. [1]

METHODS / DESIGN

System Overview

Data Collection → Model (Fuzzy NN) → Web Interface

Data Collection

- **Multiple Stocks:** AAPL, MSFT, NVDA, AMZN, TSLA, JPM, BAC, JNJ, PFE, XOM, CVX
- **Yahoo Finance:** Historical prices & fundamentals
- **FRED:** Macroeconomic indicators
- **Input:** 20-day lookback window of features
- **Output:** Predict 3-day direction (UP / DOWN)

Model Architecture

Two neural network models with identical structure. The only difference is the activation function.

Baseline Model (ReLU)	Proposed Model (Fuzzy Activation)
Input (20-day window)	Input (20-day window)
Dense (64, ReLU)	Dense (64, Fuzzy)
Dense (32, ReLU)	Dense (32, Fuzzy)
Dense (16, ReLU)	Dense (16, Fuzzy)
Output (1, Sigmoid)	Output (1, Sigmoid)

Training Setup

- Epochs: 100
- Learning Rate: 0.01
- Loss Function: Binary Cross-Entropy
- Optimizer: Adam
- Train / Test Split: 80% / 20% (temporal split)

RESULTS

Directional Accuracy (3-Day Prediction)

Fuzzy Model (Proposed)	60.9%
Baseline Model (ReLU)	53.9%
Random Chance	50.0%

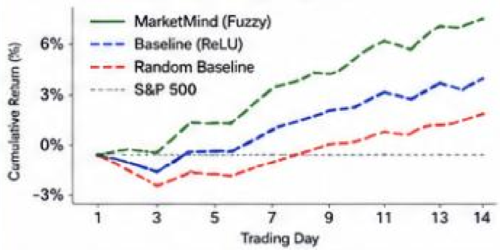
Fuzzy activation model outperformed baseline and random chance across all tested stocks.

Accuracy by Sector (Fuzzy Model)

Sector	Accuracy
Technology	~60%
Energy	~59%
Consumer	~58%
Healthcare	~62%
Financials	~58.5%

Backtesting (14-Day Simulation)

Cumulative Portfolio Return



Strategy	Return	Max Drawdown
MarketMind (Fuzzy)	+3.21%	1.03%
Baseline (ReLU)	+1.65%	1.47%
Random Baseline	+0.21%	1.92%
S&P 500	+1.02%	1.21%

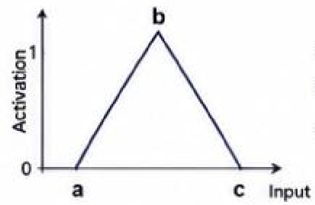
MARKETMIND DASHBOARD (PREVIEW)

MarketMind · Search tickers... · Search · Theme: Light · Portfolio

				Volume	
AAPL	Apple Inc.	$175.23	+2.34 +1.35%	Volume: 45.6M	Technology
MSFT	Microsoft Corporation	$384.56	-1.23 -0.32%	Volume: 23.5M	Technology
NVDA	NVIDIA Corporation	$825.43	+15.67 +1.93%	Volume: 67.9M	Technology

Fuzzy Activation Function

Fuzzy activation uses a triangular membership function defined by three parameters: a, b, and c.



a = start
b = peak
c = end

This allows neurons to model uncertainty and "in-between" signals more effectively than rigid activations like ReLU.

Features (Inputs)

Technical Indicators

- RSI
- MACD
- Moving Averages (50-day, 200-day)

Fundamental Indicators

- P/E Ratio (sector-adjusted)
- Earnings Growth Rate
- Debt-to-Equity Ratio

Macroeconomic Indicators

- Interest Rates
- Inflation
- Unemployment Rate
- GDP Growth Trends

IMPACT

- **Handles real market ambiguity better:** Fuzzy activations model "in-betwen" signals, leading to more realistic and stable predictions than rigid ReLU-based models.
- **Turns complex data into usable insights:** Translates volatile stock behavior into clear directional signals (guidowh) that are integrated into a simple UI.

FUTURE WORK

- Test more stocks and longer historical periods
- Improve backtesting with transaction costs
- Explore additional fuzzy activation variants
- Deploy and gather real user feedback